

NACS 645 – Quantitative Perspectives

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PROGRAM IN
NEUROSCIENCE &
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Evolution of practices in hypothesis-testing (2015-2025; non-exhaustive)

p-value

effect-sizes

probabilities

Statistics:

- t-test, ANOVA
- z-test, kolmogorov-smirnov, etc.
- Effect sizes
- Linear models, link functions
- Simulating power, sim. increasing $\{n, x\}$
- SEM, MLM (partial pooling)
- emmeans
- Comp. modelling: DDM, RL, Bayesian
- Distributions in the Bayes. domain, Bayes factor
- Bayesian estimation: MCMC, VBA

fMRI:

- GLM then t-test, ANOVA
- Increasing covariants
- 2nd-level analysis
- Model-based fMRI

A standard GLM can be written:

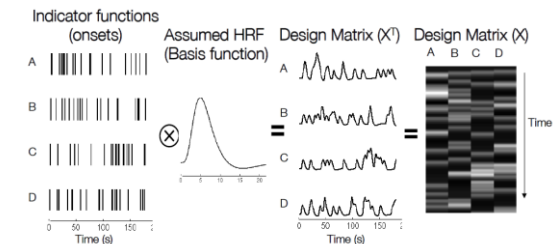
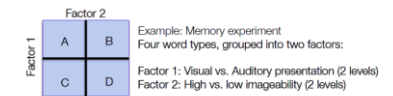
$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \quad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \mathbf{V})$$

where

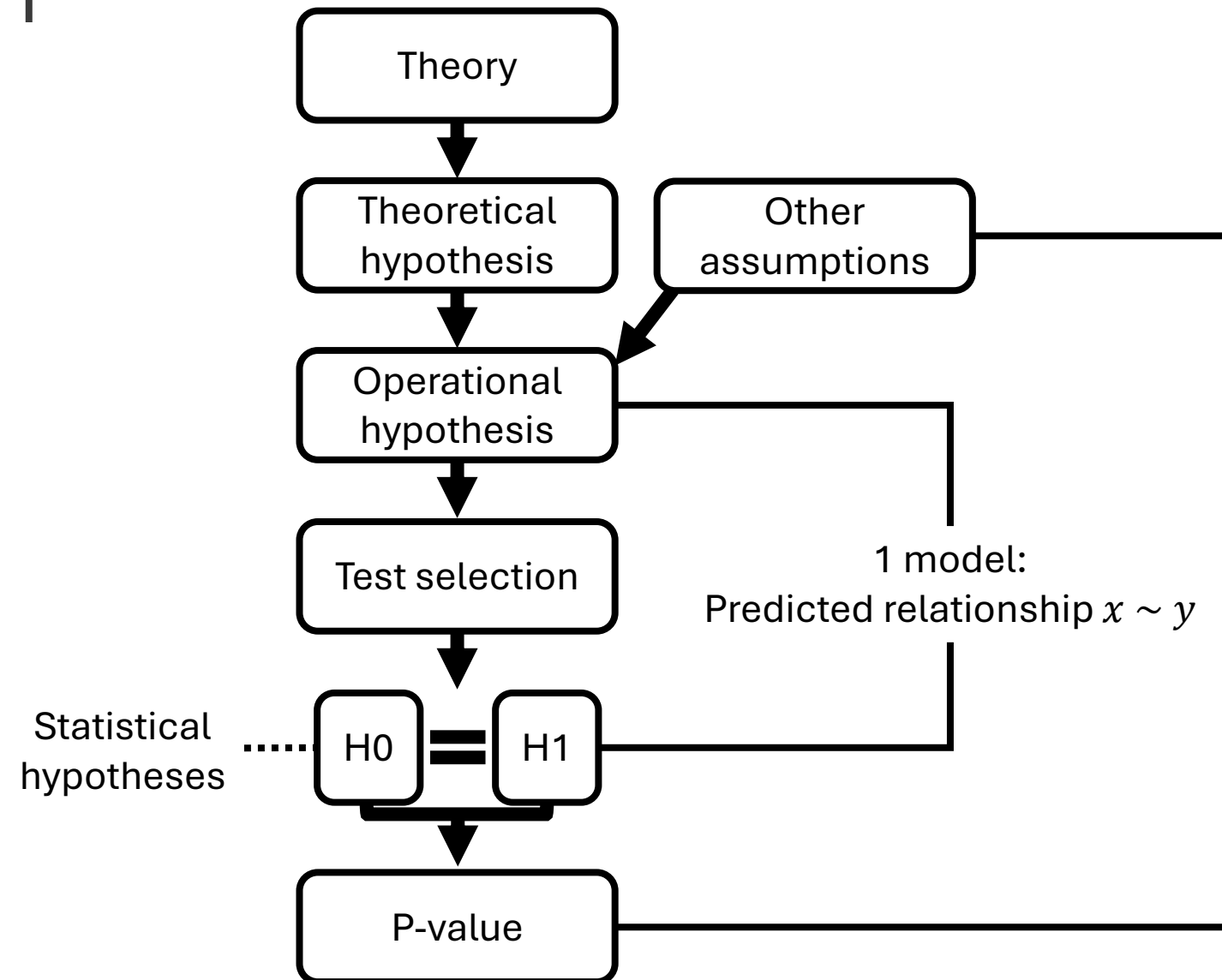
$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & X_{11} & \dots & X_{1p} \\ 1 & X_{21} & \dots & X_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & \dots & X_{np} \end{bmatrix} \times \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

fMRI Data Design matrix Regression coefficients Noise

V is the covariance matrix whose format depends on the noise model.



NHST & Reverse inference



Reverse inference: the engagement of a particular Y is inferred from finding the activation of a particular X.

Avoid the problem of reverse inference with

- > Hierarchical, model-based fMRI (verify the presence of cognitive processes first)?

- > Bayesian hypothesis testing?

- > Activation maps as priors?

Point-Null Fallacy: exactly zero-difference is virtually never true

- > Biology/social domain introduces small effects through complex causal relationships

- > Many underlying assumptions (Duhem-Quine thesis) (e.g., preprocessing)

Operational level

1. Relationship $x \sim y$
2. We predict y varies with x
3. $y \sim x_1$ and $y \sim x_2$ generate 2 different data point **distributions**

NHST

Statistical level
(test-level)

- 2 statistical hypotheses: H_0 (no difference) and H_1 (difference - **can be anything**)
- Distributions from our 2 samples come from the same population distribution (cond. 1 & 2 follow H_0)

$p < 1 - \alpha$ (i.e., $p < .05$):

- *Significant difference* = the risk to be wrong at (effect-value) is low enough. We can reject the Null Hypothesis because it is likely to be false.
- We reject H_0 ; we don't accept H_1 (**H_1 can be anything**).
We learn: it is very likely that our non-control condition doesn't follow H_0 .

$p > 1 - \alpha$:

- *No significant difference* = the risk at (effect-value) to be wrong is too high. We failed to reject H_0 because it is very not likely to be false
- We can't conclude about conditions 1 & 2 except:
We failed to reject a non-difference between cond. 1 & 2. **We learn nothing** regarding H_0 nor H_1 .

What evidence does NHST provide?

- What we can learn is restricted to our conclusions regarding H_0 for 1 model via 1 NHST test
- &
- We know nothing about how probable our results are

Null hypothesis significance testing: a review of an old and continuing controversy.

RS Nickerson - Psychological methods, 2000 - psycnet.apa.org

Null hypothesis significance testing (NHST) is arguably the most widely used approach to hypothesis evaluation among behavioral and social scientists. It is also very controversial. A ...

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